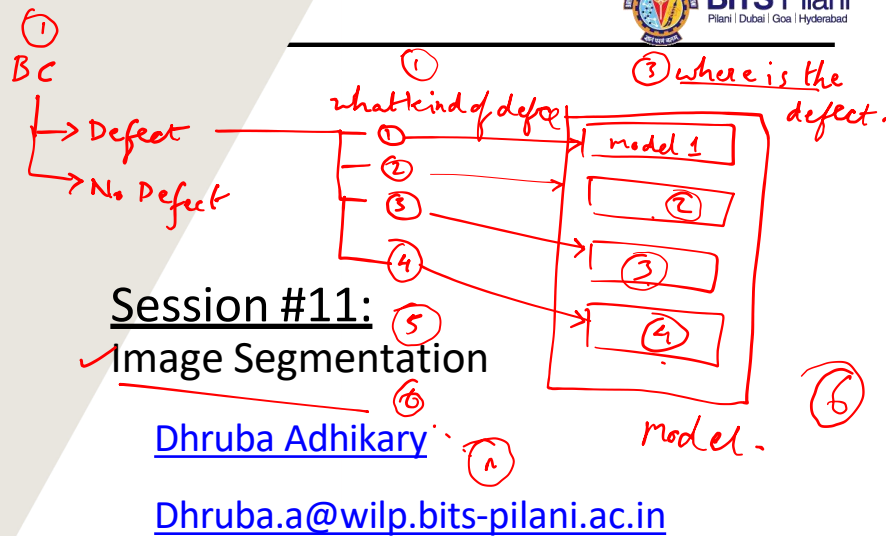




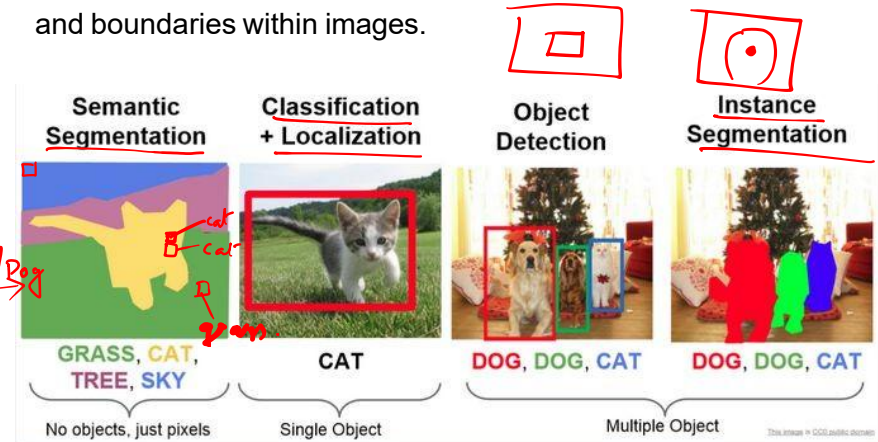
What.

→ where

Image Segmentation

CV 11: Image Segmentation

Image segmentation is a process in computer vision that involves dividing an image into multiple segments, or regions, to simplify or change the representation of an image into something more meaningful and easier to analyze. It's often used to locate objects and boundaries within images.



Acknowledgement: Slide Materials adopted from - EECS 442 – David Fouhey

CV 11: Image Segmentation

Types

Semantic Segmentation: Assigns a label to every pixel in the image, meaning each pixel is classified as belonging to a specific object class (e.g., car, tree, road).

Instance Segmentation: Similar to semantic segmentation but differentiates between different instances of the same object class (e.g., multiple cars in an image).

Panoptic Segmentation: A combination of both semantic and instance segmentation, providing a comprehensive view by segmenting both object instances and the background.

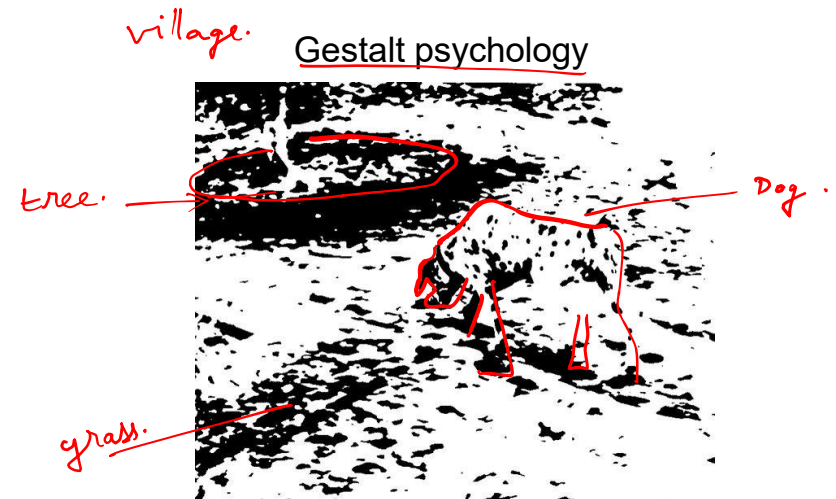
Thresholding: Simplest form where pixels are classified based on a threshold value.

Edge-based Segmentation: Uses edges (sharp changes in intensity) to identify objects.

Region-based Segmentation: Groups pixels into regions based on some predefined criteria like color, intensity, or texture.

Image Segmentation

CV 11: Image Segmentation

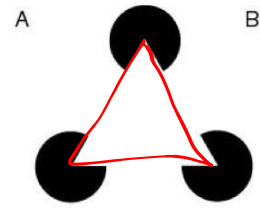


We perceive objects in its entirety before their individual parts

Gestalt: Seeing the unified whole

CV 11: Image Segmentation

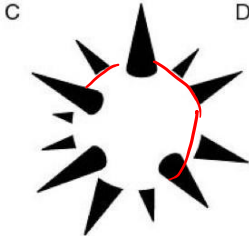
Illusory/subjective contours



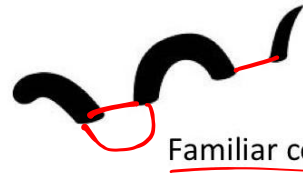
B



Occlusion



D

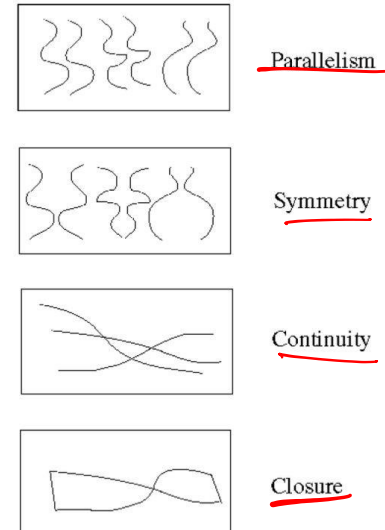
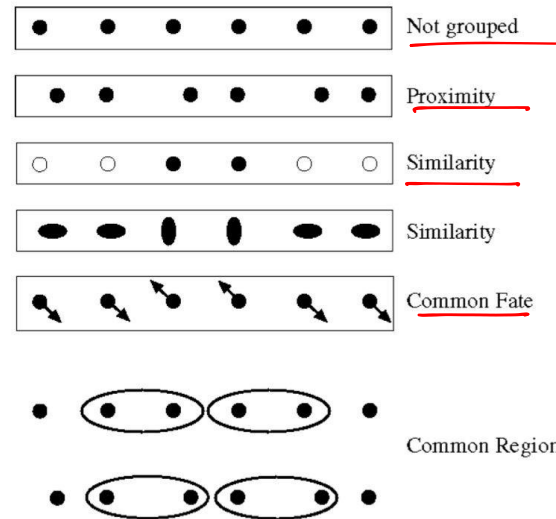


Familiar configuration

<https://ai.stanford.edu/~syyeung/cv/web/tutorial3.html>

Human perception of shape recognition

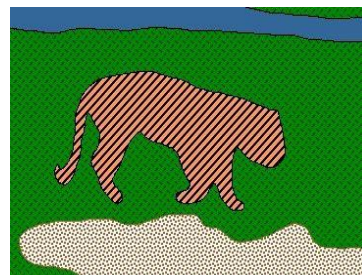
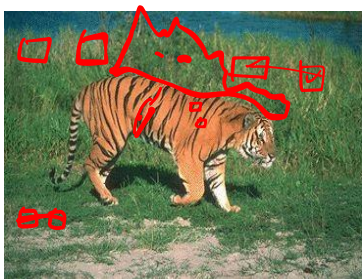
CV 11: Image Segmentation



<https://ai.stanford.edu/~syyeung/cv/web/tutorial3.html>

From images to objects

CV 11: Image Segmentation



What Defines an Object?

- Subjective problem, but has been well-studied
- Gestalt Laws seek to formalize this
 - proximity, similarity, continuation, closure, common fate
 - see [notes](#) by Steve Joordens, U. Toronto

Image Segmentation

CV 11: Image Segmentation

Intuitive to us

Hard to translate to algorithms

Strategies:

- Top Down

Pixels belong together because they come from the same object

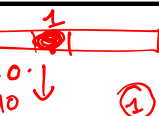
- Bottom Up

Pixels belong together because they look similar

✓


$$f[x, y] \rightarrow RGB$$
$$H_f[c] = |\{(x, y) \mid f[x, y] = c\}|$$

What do histog



- Easy to see, hard to compute

CV 11: Image Segmentation

- Break the image into K regions (segments)
- Solve this by reducing the number of colors to K and mapping each pixel to the closest color
 - photoshop demo



CV 11: Image Segmentation

- Break the image into K regions (segments)
- Solve this by reducing the number of colors to K and mapping each pixel to the closest color
 - photoshop demo



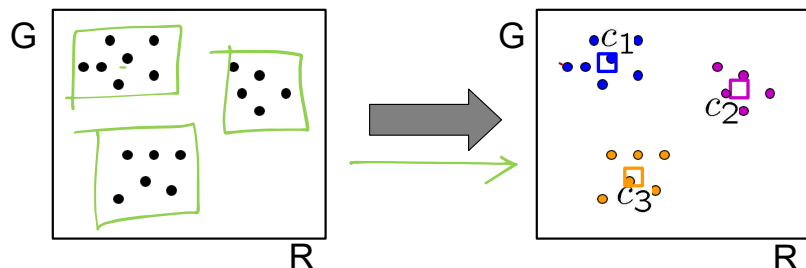
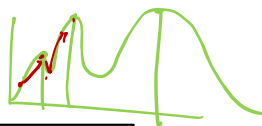
Here's what it looks like if we use two colors

Clustering

CV 11: Image Segmentation

How to choose the representative colors?

- This is a clustering problem!



Objective

- Each point should be as close as possible to a cluster center
 - Minimize sum squared distance of each point to closest center

$$\sum_{\text{clusters } i} \sum_{\text{points } p \text{ in cluster } i} \|p - c_i\|^2$$

K-means clustering

CV 11: Image Segmentation

K-means clustering algorithm

find_peaks = 3

- Randomly initialize the cluster centers, $c_1, \dots, c_K$
- Given cluster centers, determine points in each cluster
 - For each point p , find the closest c_i . Put p into cluster i
- Given points in each cluster, solve for c_i
 - Set c_i to be the mean of points in cluster i
- If c_i have changed, repeat Step 2

Demo:

<https://ai.stanford.edu/~syYeung/cvweb/tutorial3.html>

Properties

- Will always converge to *some* solution
- Can be a "local minimum"
 - does not always find the global minimum of objective function:

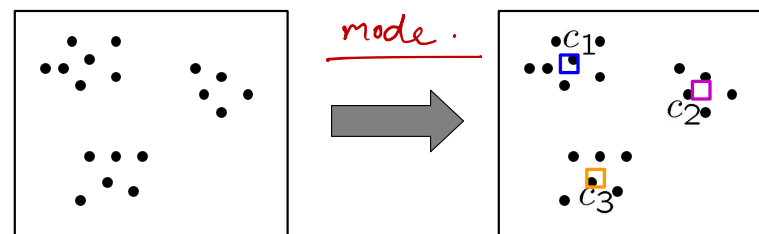
$$\sum_{\text{clusters } i} \sum_{\text{points } p \text{ in cluster } i} \|p - c_i\|^2$$

Break it down into subproblems

CV 11: Image Segmentation

Suppose I tell you the cluster centers c_i

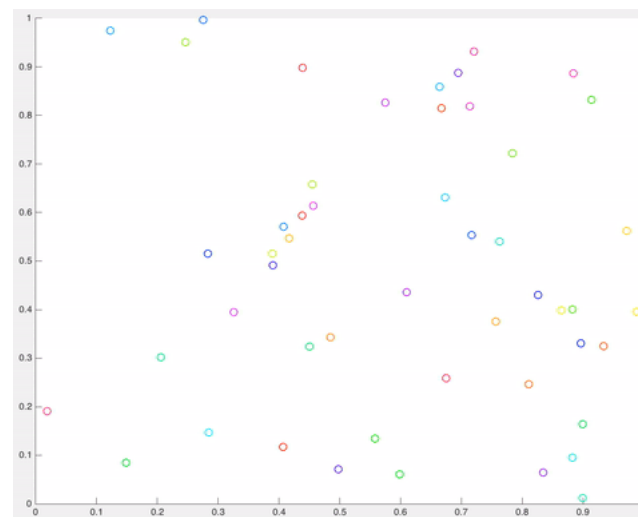
- Q: how to determine which points to associate with each c_i ?
- A: for each point p , choose closest c_i



Suppose I tell you the points in each cluster

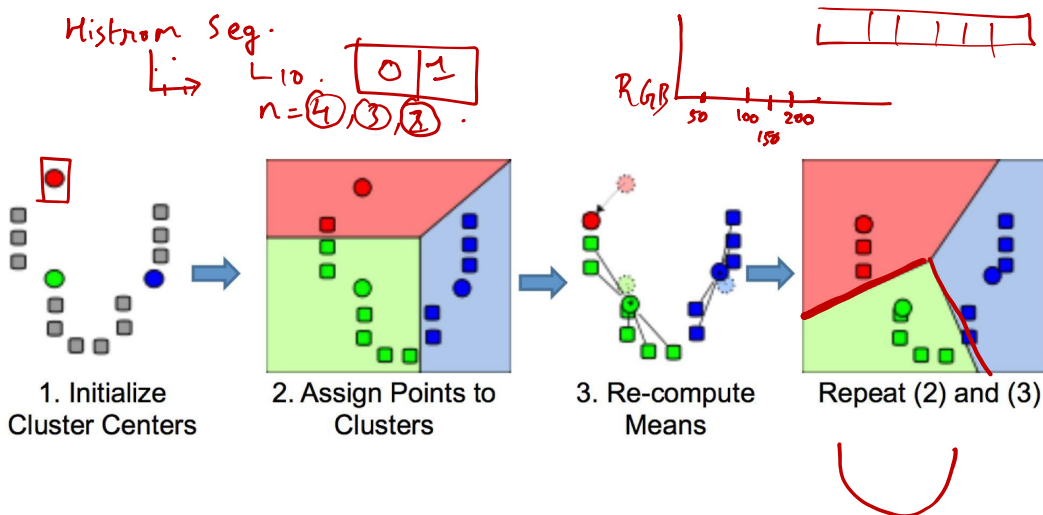
- Q: how to determine the cluster centers?
- A: choose c_i to be the mean of all points in the cluster

① Boundary.
② intensity based
prone to mistake.



K means clustering

CV 11: Image Segmentation



Probabilistic clustering

CV 11: Image Segmentation

① shape of the boundary

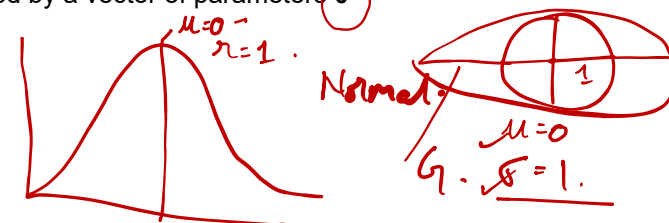
Basic questions

- what's the probability that a point x is in cluster m ?
- what's the shape of each cluster?

K-means doesn't answer these questions

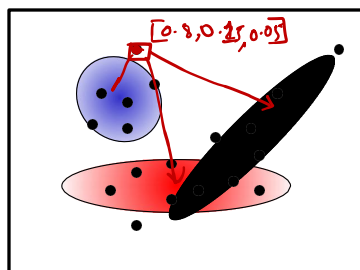
Basic idea

- instead of treating the data as a bunch of points, assume that they are all generated by sampling a continuous function
- This function is called a **generative model**
 - defined by a vector of parameters θ



Mixture of Gaussians

CV 11: Image Segmentation



$\mu=0.01, \sigma=1$
 $0.1, 1$
 $0, 2, 0.5$

One generative model is a mixture of Gaussians (MOG)

- K Gaussian blobs with means μ_b covariance matrices V_b , dimension d

– blob b defined by: $P(x|\mu_b, V_b) = \frac{1}{\sqrt{(2\pi)^d |V_b|}} e^{-\frac{1}{2}(x-\mu_b)^T V_b^{-1}(x-\mu_b)}$

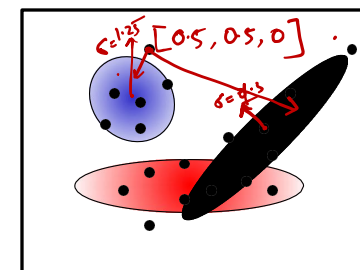
- blob b is selected with probability α_b
- the likelihood of observing x is a weighted mixture of Gaussians

$$P(x|\theta) = \sum_{b=1}^K \alpha_b P(x|\theta_b)$$

- where $\theta = [\mu_1, \dots, \mu_n, V_1, \dots, V_n]$

Expectation maximization (EM)

CV 11: Image Segmentation



$\mathcal{E} (0.6, 0.35, 0.05)$
 $\downarrow$
 $\mu \rightarrow \mu, \sigma$
 $\downarrow$
 $(0.65, 0.3, 0.05)$
 $\downarrow$
 μ_1, σ_1
 $\downarrow$
 $(0.9, 0.1, 0)$

Goal

- find blob parameters θ that maximize the likelihood function:

$$P(\text{data}|\theta) = \prod_x P(x|\theta)$$

Approach:

1. **E step:** given current guess of blobs, compute ownership of each point
2. **M step:** given ownership probabilities, update blobs to maximize likelihood function
3. repeat until convergence

E-step

- compute probability that point x is in blob i , given current guess of θ

$$P(b|x, \mu_b, V_b) = \frac{\alpha_b P(x|\mu_b, V_b)}{\sum_{i=1}^K \alpha_i P(x|\mu_i, V_i)}$$

0.99 0 0 0.11



M-step

- compute probability that blob b is selected

$$\alpha_b^{new} = \frac{1}{N} \sum_{i=1}^N P(b|x_i, \mu_b, V_b) \quad N \text{ data points}$$

- mean of blob b

$$\mu_b^{new} = \frac{\sum_{i=1}^N x_i P(b|x_i, \mu_b, V_b)}{\sum_{i=1}^N P(b|x_i, \mu_b, V_b)}$$

- covariance of blob b

$$V_b^{new} = \frac{\sum_{i=1}^N (x_i - \mu_b^{new})(x_i - \mu_b^{new})^T P(b|x_i, \mu_b, V_b)}{\sum_{i=1}^N P(b|x_i, \mu_b, V_b)}$$

Applications of EM

Turns out this is useful for all sorts of problems

- any clustering problem
- any model estimation problem
- missing data problems
- finding outliers
- segmentation problems
 - segmentation based on color
 - segmentation based on motion
 - foreground/background separation
- ...

<https://github.com/annoviko/pyclustering>

Problems with EM

Local minima



Need to know number of segments

Need to choose generative model

→ Bayesian Model.



Mean-shift for image segmentation

Useful to take into account spatial information

- instead of (R, G, B), run in (R, G, B, x, y) space
- D. Comaniciu, P. Meer, Mean shift analysis and applications, 7th International Conference on Computer Vision, Kerkyra, Greece, September 1999, 1197-1203.

95, 98.
hit
- <http://www.caip.rutgers.edu/riul/research/papers/pdf/spatmsft.pdf>
(255, 10, 10, 1, 0).

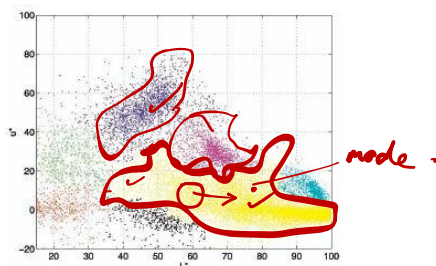
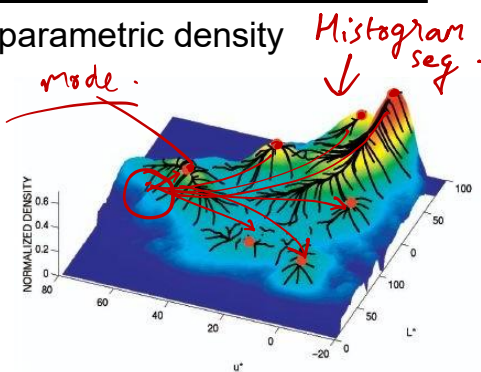


More Examples: http://www.caip.rutgers.edu/~comanici/segm_images.html

Mean shift algorithm

CV 11: Image Segmentation

Try to find *modes* of this non-parametric density

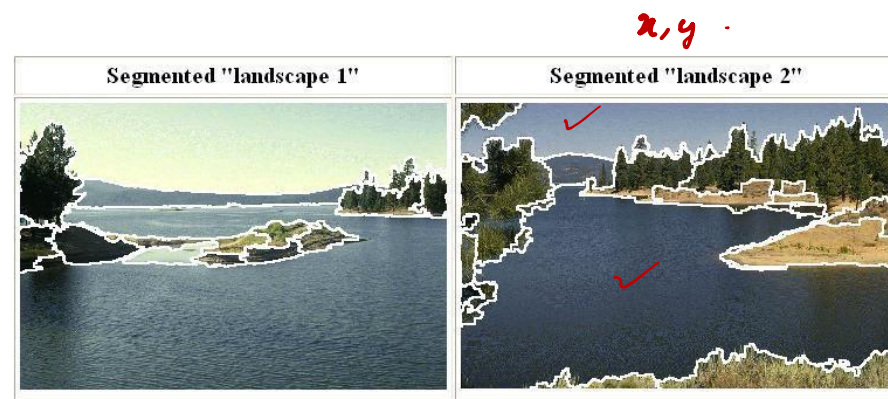


Mean shift segmentation

CV 11: Image Segmentation

D. Comaniciu and P. Meer, Mean Shift: A Robust Approach toward Feature Space Analysis, PAMI 2002.

Versatile technique for clustering-based segmentation



Kernel density estimation

CV 11: Image Segmentation

Kernel density estimation function

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^n K\left(\frac{x - x_i}{h}\right)$$

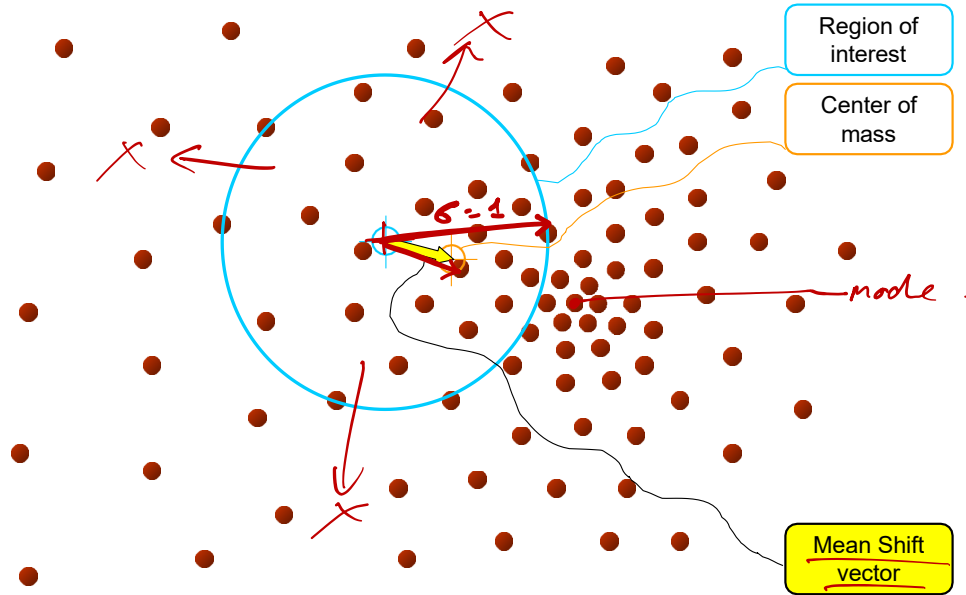


Gaussian kernel

$$K\left(\frac{x - x_i}{h}\right) = \frac{1}{\sqrt{2\pi}} e^{-\frac{(x - x_i)^2}{2h^2}}$$

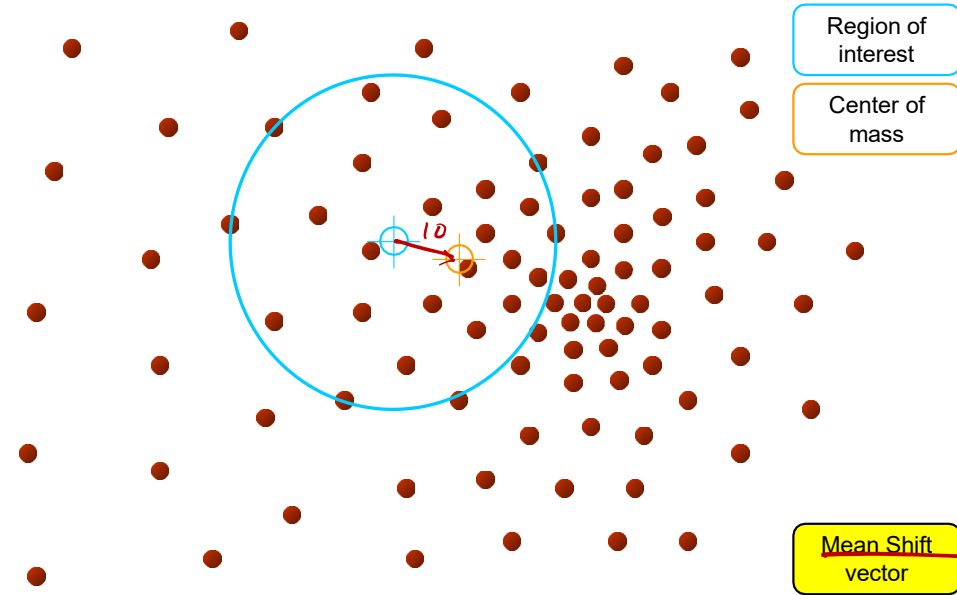


Mean shift



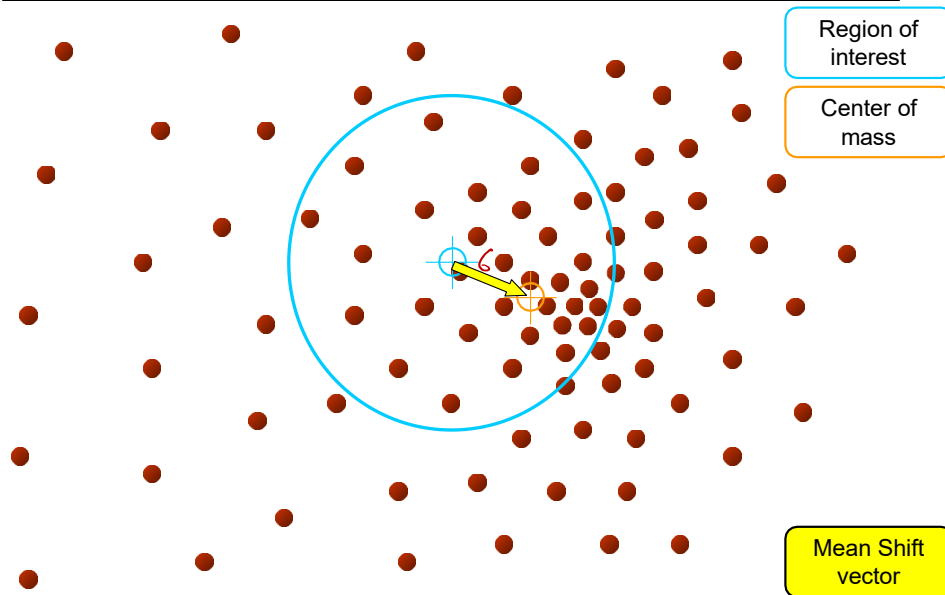
Slide by Y. Ukrainitz & B. Sarel

Mean shift



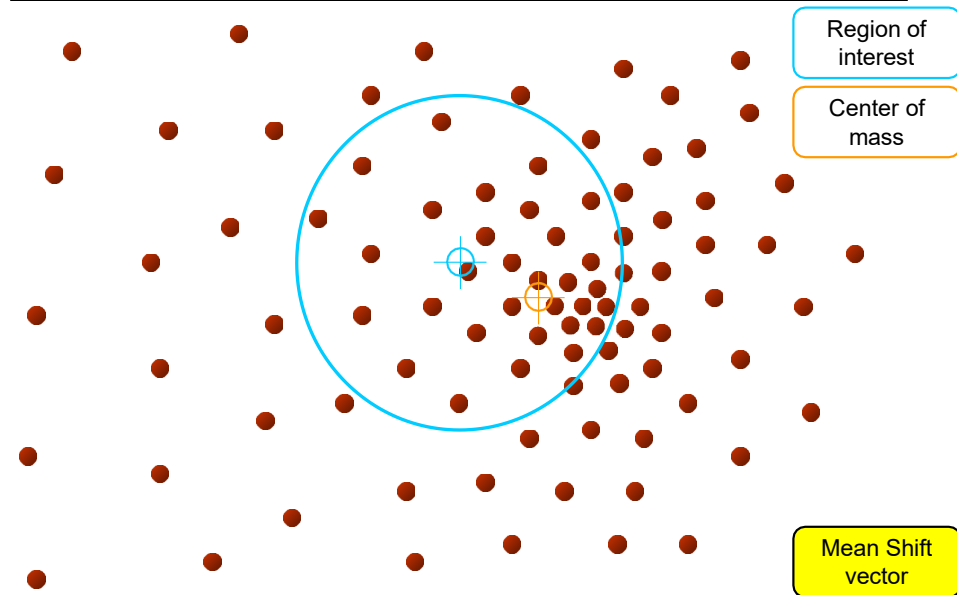
Slide by Y. Ukrainitz & B. Sarel

Mean shift



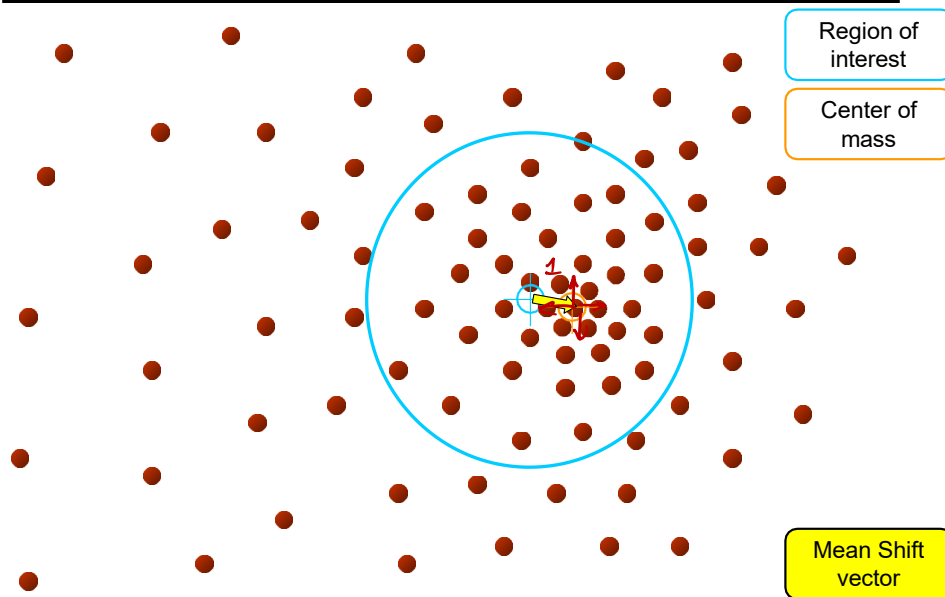
Slide by Y. Ukrainitz & B. Sarel

Mean shift



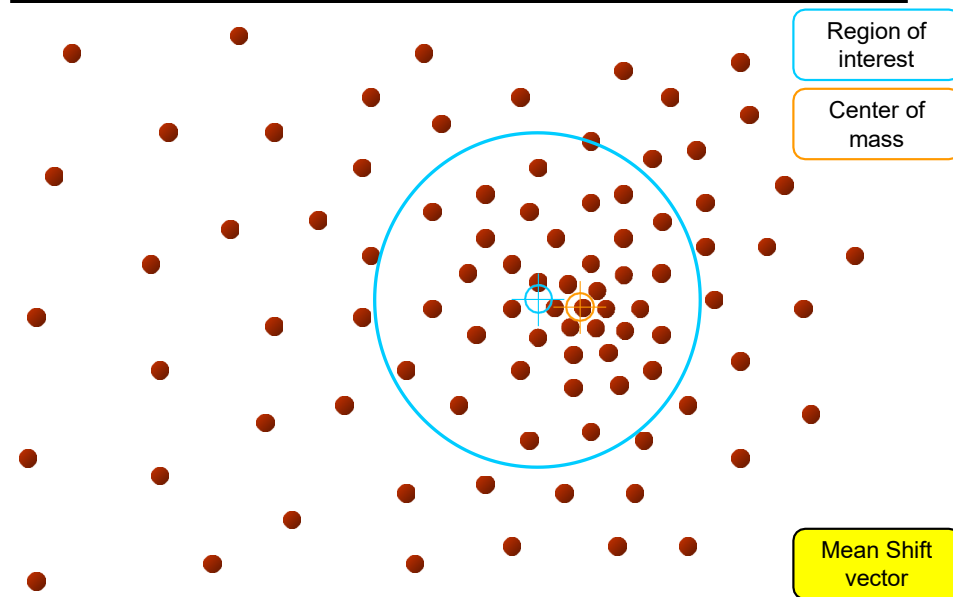
Slide by Y. Ukrainitz & B. Sarel

Mean shift



Slide by Y. Ukrainitz & B. Sarel

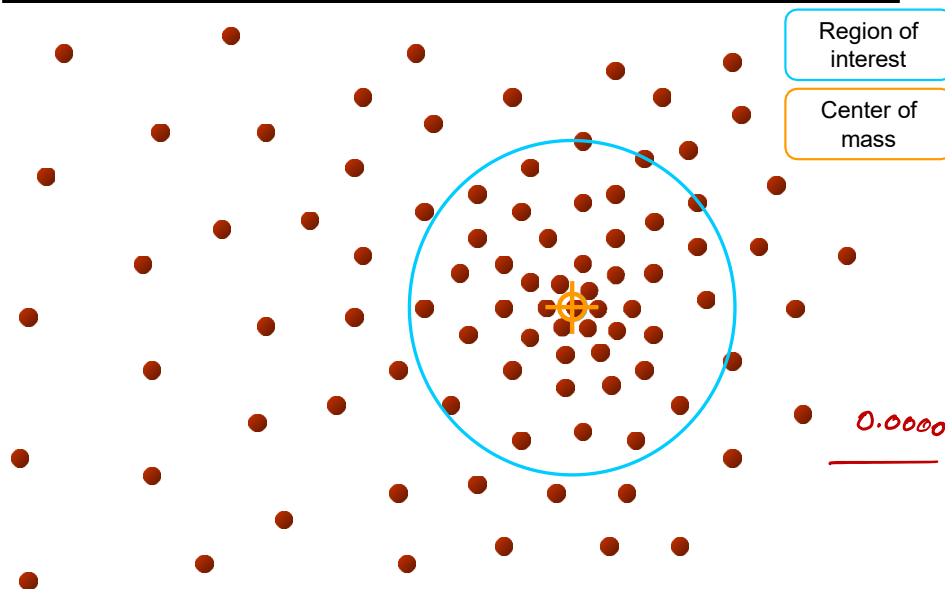
Mean shift



Slide by Y. Ukrainitz & B. Sarel

Mean shift

CV 11: Image Segmentation



Slide by Y. Ukrainitz & B. Sarel

Computing the Mean Shift

CV 11: Image Segmentation

Simple Mean Shift procedure:

- Compute mean shift vector
- Translate the Kernel window by $\mathbf{m}(\mathbf{x})$

Handwritten notes:

KDE = $\frac{1}{n} \sum_{i=1}^n \frac{1}{h} \exp\left(-\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{h^2}\right)$

Simple Mean Shift procedure:

- Compute mean shift vector
- Translate the Kernel window by $\mathbf{m}(\mathbf{x})$

Diagram illustrating the computation of the mean shift vector $\mathbf{m}(\mathbf{x})$. The formula is shown as:

$$\mathbf{m}(\mathbf{x}) = \frac{\sum_{i=1}^n \mathbf{x}_i g\left(\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{h^2}\right)}{\sum_{i=1}^n g\left(\frac{\|\mathbf{x} - \mathbf{x}_i\|^2}{h^2}\right)} - \mathbf{x}$$

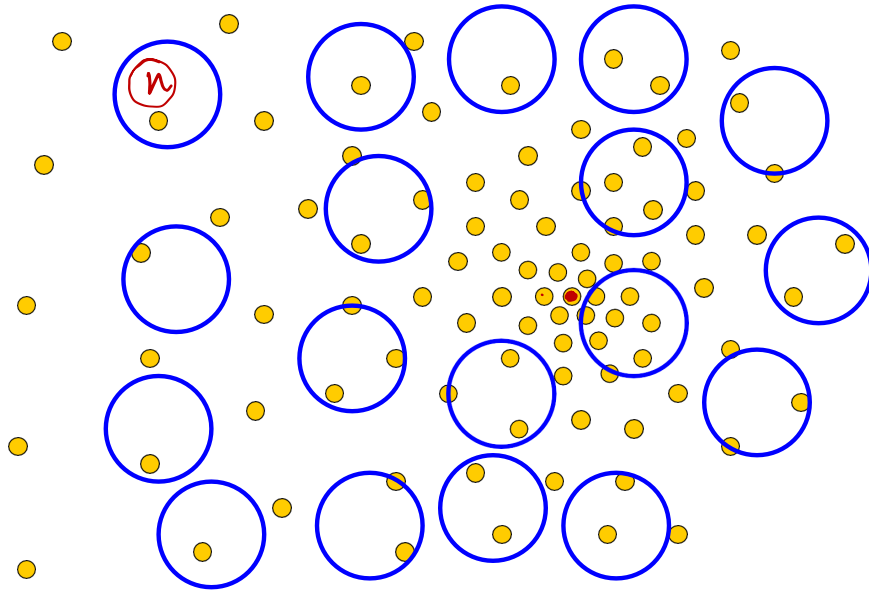
The diagram shows a blue circle (Region of interest) and a cluster of brown dots. A red arrow points from the center of mass (orange dot) to the center of the blue circle, labeled $\mathbf{m}(\mathbf{x})$. A red arrow points from the center of the blue circle to the center of mass, labeled $\mathbf{x}$. A red arrow points from the center of the blue circle to the center of mass, labeled $\mathbf{m}(\mathbf{x})$.

Slide by Y. Ukrainitz & B. Sarel

$$g(\mathbf{x}) = -k'(\mathbf{x})$$

Real Modality Analysis

CV 11: Image Segmentation

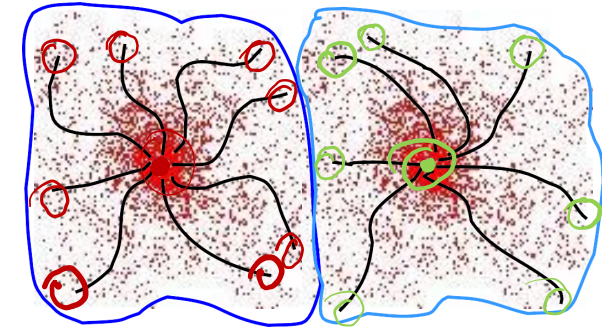


Attraction basin

CV 11: Image Segmentation

Attraction basin: the region for which all trajectories lead to the same mode

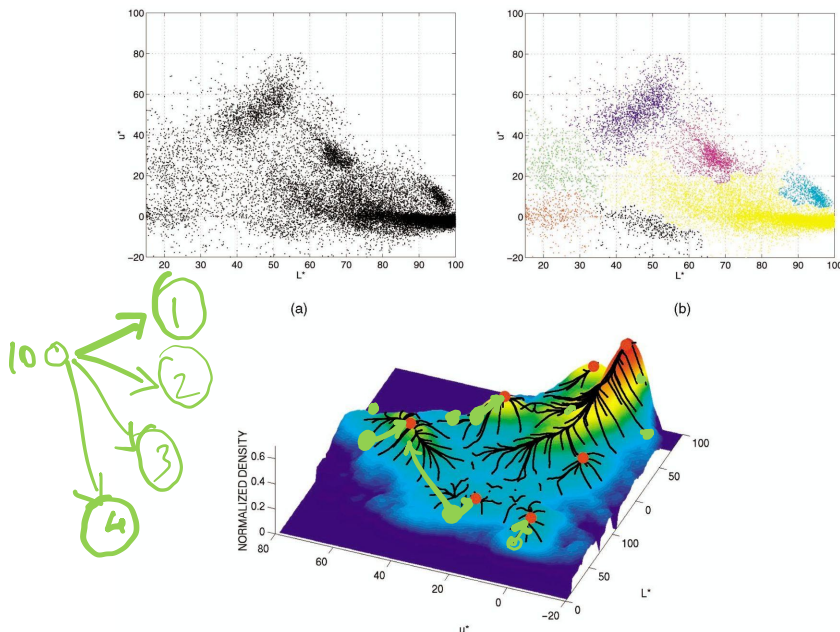
Cluster: all data points in the attraction basin of a mode



Slide by Y. Ukrainitz & B. Sarel

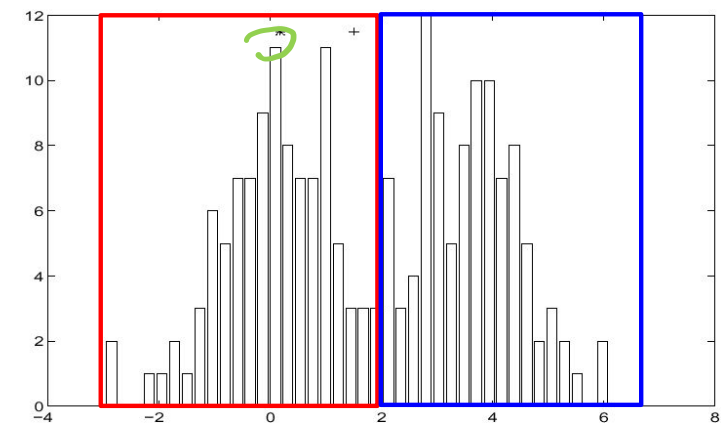
Attraction basin

CV 11: Image Segmentation



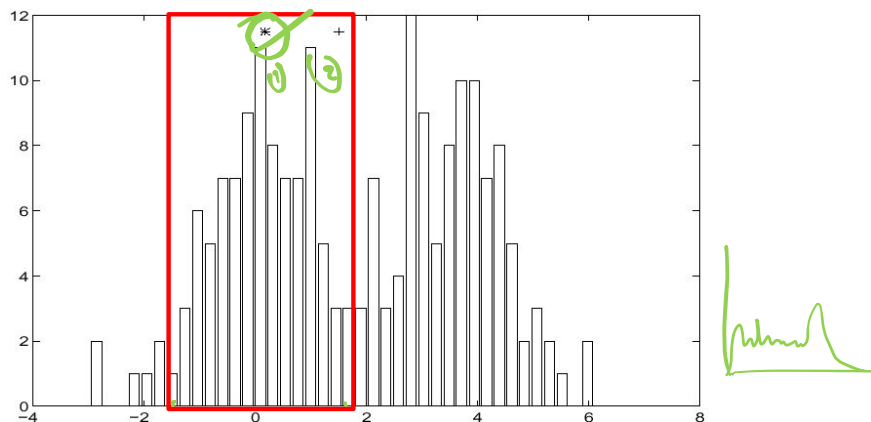
Finding Modes in a Histogram

CV 11: Image Segmentation



How Many Modes Are There?

- Easy to see, hard to compute



Iterative Mode Search

1. Initialize random seed, and window W
2. Calculate center of gravity (the "mean") of W : $\sum_{x \in W} xH(x)$
3. Translate the search window to the mean
4. Repeat Step 2 until convergence

Mean shift clustering

The mean shift algorithm seeks *modes* of the given set of points

1. Choose kernel and bandwidth
2. For each point:
 - a) Center a window on that point
 - b) Compute the mean of the data in the search window
 - c) Center the search window at the new mean location
 - d) Repeat (b,c) until convergence
3. Assign points that lead to nearby modes to the same cluster

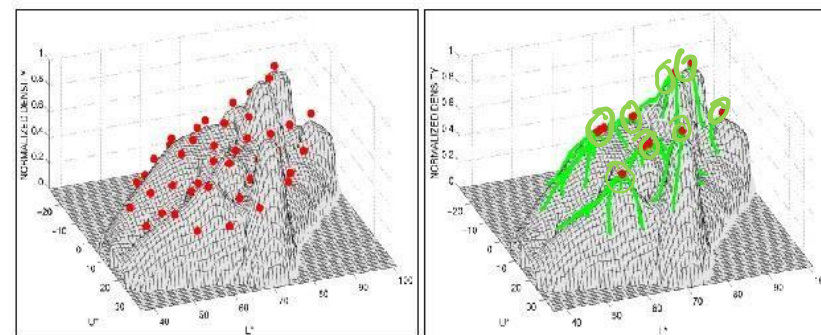


Mean-Shift

Approach

MS + KMC
+ PC

- Initialize a window around each point
- See where it shifts—this determines which segment it's in
- Multiple points will shift to the same segment



Mean shift trajectories

Mean shift segmentation results





Mean-shift: other issues

Speedups

- Binned estimation
- Fast search of neighbors KDE
- Update each window in each iteration (faster convergence)

$$\frac{MS + KM}{+ PG}$$

Other tricks

- Use kNN to determine window sizes adaptively

Lots of theoretical support

D. Comaniciu and P. Meer, Mean Shift: A Robust Approach toward Feature Space Analysis, PAMI 2002.

<http://www.caip.rutgers.edu/~comanici/MSPAMI/msPamiResults.html>

Mean shift pros and cons

Pros

- Good general-practice segmentation
- Flexible in number and shape of regions
- Robust to outliers

Cons

- Have to choose kernel size in advance
- Not suitable for high-dimensional features

When to use it

- Oversegmentation
- Multiple segmentations
- Tracking, clustering, filtering applications

